

TECHNICAL REPORT

MTA Daily Ridership

COVID-19 Recovery Analysis

A comprehensive data analysis of how New York's transit system was impacted by COVID-19 and how each mode has recovered.

■ Data Period: March 2020 – October 2024

■ 1,706 days tracked

■ 7 transit modes

DEPI Data Analysis Track · CodeAlpha Internship

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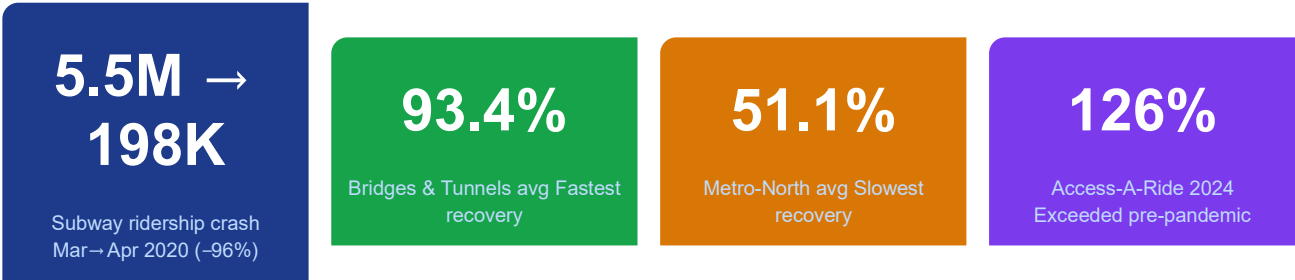
1. Executive Summary

COVID-19 hit New York's public transit harder than almost any event in its history. In just five weeks in early 2020, subway ridership collapsed by **96%** — from 5.5 million daily riders to fewer than 200,000. Since then, recovery has been uneven and structurally significant: some modes fully rebounded while

others stabilized at permanently lower baselines, reflecting deep behavioral shifts in how New Yorkers live, work, and move.

This report presents a full data analysis of MTA ridership from **March 2020 to October 2024** across 7 transit modes, examining the collapse, the recovery trajectory, and the long-term implications for transit planning in a post-pandemic city.

"COVID didn't just temporarily disrupt transit — it accelerated behavioral changes in remote work, car usage, and paratransit demand that were already quietly underway."



2. Dataset Overview & Business Constraints

2.1 About the Dataset

The dataset is sourced from the **MTA Open Data Portal** and covers daily ridership for all major New York City transit modes from March 1, 2020 through October 31, 2024 — a total of 1,706 calendar days.

Column	Description	Type	Note
Date	Calendar date of observation	Date	Daily granularity
Subways: Total Estimated Ridership	Estimated daily subway riders	Integer	Core metric
Subways: % of Comparable Pre-Pandemic Day	Recovery % vs equivalent 2019 day	Integer (0–143)	■ Keep as int
Buses: Total Estimated Ridership	Daily bus passengers	Integer	—
Buses: % of Comparable Pre-Pandemic	Day Recovery %	Integer	■ Keep as int
LIRR / Metro-North / Access-A-Ride	Same structure per mode	Integer	—
Bridges and Tunnels: Total Traffic	Vehicle crossings per day	Integer	Not riders — vehicles
Staten Island Railway: Total Estimated Ridership	Smallest mode	Integer	—

2.2 Business Constraints & Analytical Challenges

Before analysis began, the team identified several real-world constraints that shaped how the data was interpreted and what questions could meaningfully be answered:

Integer % Columns — Correct by Design

The "% of Pre-Pandemic" columns store whole numbers (e.g. 97, 143) — not decimals. This is intentional: the values represent percentage points, not ratios. Converting them to float (0.97) would break all percentage-based visuals. **Decision: keep as int, apply % label in formatting only.**

Ridership ≠ Vehicle Traffic

Bridges & Tunnels measures **vehicle crossings**, not passengers. This makes direct comparisons with other modes misleading. In Power BI, this required separate axis scaling and clear labeling to prevent misinterpretation.

Estimated Ridership — Not Exact Counts

All ridership figures are **estimates** based on MetroCard swipes and OMNY taps — they exclude fare evaders, paratransit no-shows, and irregular entries. This limits precision but is sufficient for trend analysis.

Day-of-Week Comparability

Pre-pandemic % is calculated by comparing each day to the **same day of week** in the same calendar period of 2019. This controls for weekend effects, holidays, and seasonal variation — but means a "100%" value on a quiet Sunday ≠ 100% on a busy Monday.

Data Coverage Gap — Bus Ridership April–August 2020

Bus ridership data shows near-zero values (under 6,000) for April–July 2020, then jumps to ~935,000 in September. This appears to be a **reporting methodology change** by MTA, not an actual ridership anomaly. These months were excluded from bus trend analysis.

No External Variables in Dataset

The dataset contains only ridership counts — no employment data, remote work rates, weather, or policy events. All behavioral interpretations (WFH effect, car preference) are **inferred from patterns**, not directly measurable from this data alone.

3. What We Built in Power BI

The Power BI dashboard was designed to make 1,706 days of transit data instantly navigable for a non-technical audience — giving decision-makers a clear view of recovery progress across all 7 transit modes without requiring any data background.

3.1 Data Model & Preparation

Step 1 — Data Import: Loaded the MTA CSV directly into Power BI via Get Data → Text/CSV. The date column was automatically parsed as a Date type.

Step 2 — Calendar Table: Created a Date table using CALENDAR() DAX to enable proper time intelligence (year-over-year, month filtering via slicers).

Step 3 — Column Typing: All ridership columns kept as Whole Number. All % columns kept as Whole Number with % display format applied at the visual level — NOT converted to decimal, to avoid the ×100 multiplication error.

Step 4 — Calculated Columns: Added Year and Month Name columns using YEAR([Date]) and FORMAT([Date], "MMMM") for time-based slicing.

Step 5 — Measures: Created DAX measures for average ridership, max/min values, and conditional recovery status labels (Recovered / Partial / Struggling).

3.2 Key DAX Measures Used

Measure Name	DAX Formula	Purpose
Avg Subway Ridership	AVERAGE([Subways: Total Estimated Ridership])	KPI card
Max Ridership Day	MAXX(ALL(MTA), [Subways: Total Estimated Ridership])	Peak insight
Recovery Status	IF([Avg % Pre-Pandemic] >= 90, "Recovered", IF([Avg % Pre-Pandemic] >= 60, "Partial", "Struggling"))	Status labeling
YoY Change %	DIVIDE([Avg This Year] - [Avg Last Year], [Avg Last Year])	Yearly trend

3.3 Dashboard Pages & Visuals

■ Overview Page

- KPI Cards: Peak subway day (5.5M), Lowest day (198K), B&T; recovery (93.4%), Metro-North (51.1%)
- Bar Chart: Avg % of pre-pandemic per mode — sorted descending
- Line Chart: Subway ridership trend 2020–2024 with date slicer
- Insight Text Cards: 4 key findings with icons

■ Trend Analysis Page

- Multi-line chart: All 7 modes % recovery on same axis — shows divergence clearly
- Mode selector slicer for filtering to individual transit types
- Clustered Bar: Year-over-year ridership by mode (2020–2024)
- 100% Reference Line added as constant line to show full recovery threshold

■ Recovery Scorecard Page

- Table visual with conditional formatting: Green (>90%), Yellow (60-90%), Red (<60%)
- Gauge visual: Subway recovery at 71% vs 100% target
- Stacked area chart: Recovery accumulation over 4 years
- Summary text: 3 key behavioral insights derived from patterns

4. Key Findings & Insights

The data analysis produced 4 major findings, each supported by specific patterns in the ridership dataset:

Finding 1: Cars Won the Pandemic

Bridges & Tunnels — representing personal vehicle traffic — recovered to **93.4% average** across the full period, while subways averaged only 55.5%. The gap in 2020 alone was 42 percentage points (68% vs 26%). When given a choice, New Yorkers consistently chose driving over shared transit. By 2024, B&T, exceeded 100% of pre-pandemic levels while subways remained at 71%.

Finding 2: Remote Work Created a New Permanent Baseline

Metro-North — which carries suburban commuters — averaged only **51.1%** of pre-pandemic levels across the full period. This is not a slow recovery: it is a new structural equilibrium. Hybrid work schedules (2–3 days in office) have become permanent for suburban professionals. The data shows Metro-North stopped recovering around 2023 and stabilized, suggesting this IS the new normal.

Finding 3: Paratransit Demand Surged Beyond Pre-Pandemic

Access-A-Ride — serving mobility-limited riders — is the only mode that **exceeded** 100% of pre-pandemic levels, reaching 126% by 2024. This reflects three factors: NYC's aging population, expanded eligibility criteria post-COVID, and pent-up demand from the period when services were restricted. This is a planning challenge, not just good news.

Finding 4: Subway Recovery Plateaued at ~71%

Subway ridership recovered steadily from 2020 to 2023, then **plateaued around 70–71%** of pre-pandemic levels. Tourism-driven and essential worker trips came back; daily white-collar office commutes did not fully return. The 29% gap is not a recovery lag — it reflects a fundamentally changed commuting culture where millions of office workers now commute 3 days a week instead of 5.

4.1 Recovery Scorecard by Mode

Transit Mode	2020 Avg %	2021 Avg %	2022 Avg %	2023 Avg %	2024 Level	Status
■ Bridges & Tunnels	68%	87%	97%	100%	~101%	Fully Recovered
■ Access-A-Ride	55%	72%	87%	98%	~126%	Exceeded Baseline
■ LIRR	22%	43%	64%	72%	~74%	Strong Recovery
■ Subways	26%	46%	61%	70%	~71%	Partial Recovery
■ Buses	29%	48%	60%	69%	~69%	Partial Recovery
■ Metro-North	18%	38%	57%	65%	~67%	Stabilized (Low)
■ Staten Island Rwy	19%	31%	41%	48%	~50%	Significantly Below

5. Conclusions & Strategic Implications

The MTA isn't just recovering from COVID — it is adapting to a post-COVID city that looks structurally different from the one that existed in February 2020. Three things stand out:

The Office Commute May Never Fully Return

Metro-North and LIRR data shows clearly that the Monday-to-Friday, 9-to-5 commute pattern has changed permanently. Hybrid work is not a temporary phase — it is the new contract between employers and suburban workers. Transit planning needs to model for a 3-day commute week, not 5.

Local & Essential Trips Are the New Backbone

Subway and bus ridership at ~70% is being driven by essential workers, students, and local trips — not white-collar office commuters. This is a more diverse and potentially more recession-resistant ridership base than the old commuter-heavy model. The MTA should invest in reliability and off-peak service for this audience.

Paratransit Is a Growing Obligation

Access-A-Ride exceeding 100% capacity is both good news and a planning challenge. As NYC's population ages and eligibility expands, paratransit demand will keep growing. Proactive investment in this service is not optional — it is a civil rights and mobility equity obligation.

6. Data Notes & Methodology

This analysis uses the MTA's publicly available Daily Ridership dataset from the MTA Open Data Portal. Recovery percentages compare each day's ridership to the equivalent pre-pandemic day (same day of week, same seasonal period, baseline: early 2020 before COVID restrictions). An average of 100% indicates full recovery; above 100% means demand exceeded the pre-pandemic baseline. Bus ridership anomalies in April–August 2020 are attributed to a reporting methodology change by MTA and should be interpreted with caution.

Tools used: Python (pandas, reportlab, openpyxl) for data cleaning and file generation; Power BI for interactive dashboards and DAX measures; Excel for pivot table analysis; HTML/JavaScript for web-based dashboard delivery.